

Contents

Chapter	Page
1 General	
Installation Survey, Special Tools	1.1/1
Overall View	1.1/1
Technical Data	1.2/1
Functional Descriptions	1.3/1
Technical Safety and Operating Regulations	1.4/1
Troubleshooting	1.5/1
2 Removal and Installation of Switch Panel	2.1/1
3 Removal and Installation of Control Unit	3.1/1
4 Renewing Relay for Float Switch	4.1/1
5 Removal and Installation of Float Switch	5.1/1
6 Renewing Glow Plug, Checking Function	6.1/1
7 Renewing Series Resistor for Ignition Spark Generator	7.1/1
8 Renewing Plug Resistors	8.1/1
9 Removal and Installation of Thermal Switch	9.1/1
10 Removal and Installation of Suppressor Combination	10.1/1
11 Removal and Installation of Ignition Spark Generator	11.1/1
12 Removal and Installation of Overheating Switch	12.1/1
13 Removal and Installation of Temperature Sensor	13.1/1
14 Removal and Installation of Fuel Feed Pump	14.1/1
15 Renewing Solenoid Valve for Combustion Air	15.1/1
16 Removal and Installation of Heater Unit	16.1/1
17 Renewing Gasket for Exhaust Pipe	17.1/1
18 Renewing Impeller	18.1/1
19 Renewing Electric Motor	19.1/1
20 Renewing Heat Exchanger and Burner	20.1/1
21 Renewing Cable Harness	21.1/1
22 Checking CO Value and CO² Value in Interior of Box-Type Body	22.1/1

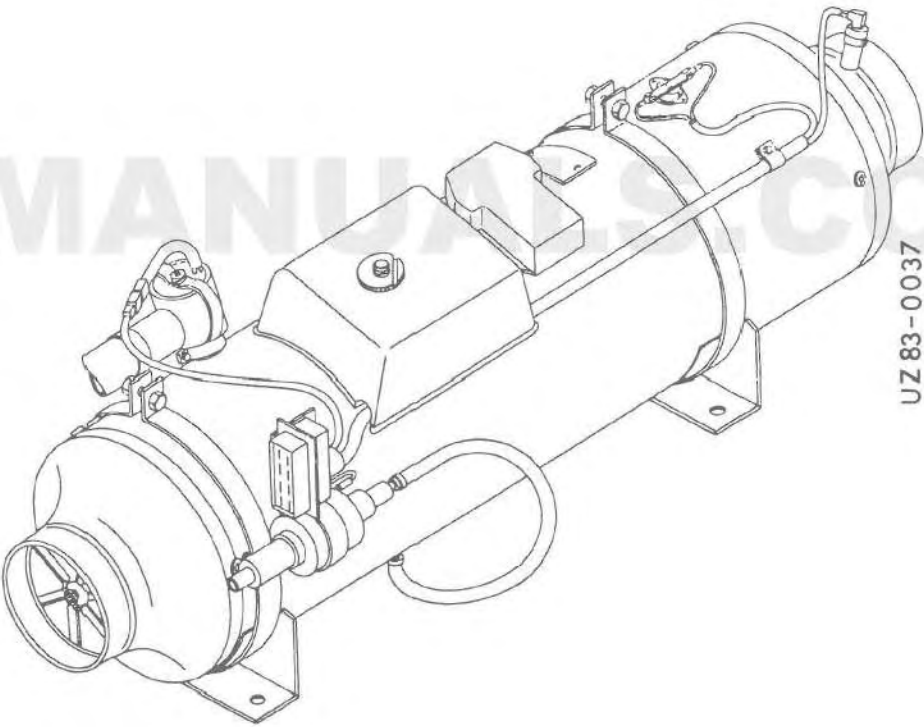
Installation Survey

Chassis		Auxiliary heater		
Model	Sales designation	Manufacturer	Type	Installation
435.115	U 1300 L	Messrs. Eberspächer	V 7 S	in box-type body

Special tool– Messrs. Eberspächer

Cons. No.	Designation	Special tool
1	Socket wrench (for glow plug)	2512 4405 00 10

Overall view



Technical Data

Auxiliary Heater Eberspächer V 7 S

Heating capacity $\pm 10\%$		kW	3 to 12
Fuel			Diesel fuel
Fuel consumption at 3 kW (pump frequency 2.5 Hz) at 12 kW (pump frequency 8.5 Hz)		l/h	0,4 1,3
Rated voltage		Volt	24
Weight approx.		Kp	21
power consumption Start Continuous duty $\pm 10\%$	approx.	Watt Watt	420 110
Permissible exhaust back pressure approx.		mm water column	10 - 5
Permissible exhaust underpressure approx.		mm water column	
Volume of heated air approx. (at rated voltage, measured with run - in nozzle $\varnothing 100$ without back pressure)		Kg/h	360
Permissible temperature of combustion air taken in		° C	- 31 to+ 30
ABG test symbol			~ S 130
Inspection values at rated voltage: - Thermal switch after running time - Response time safety switch	max. max.	sec. sec.	180 180

Functional Description

GENERAL

The auxiliary heater can be operated independent of the vehicle engine and serves the purpose of ventilating and heating the box-type body as well as preheating the auxiliary batteries at extremely low temperatures.

The heater is operated with fresh air and is installed in the box-type body in a separate compartment under the bench seat. The fresh air is obtained via an opening protected against splashwater in the centre of the front wall between the box-type body and the driver's cab.

The warm air is directed via a flexible pipe from the heater to an air duct on the left side wall. This duct has outflow openings distributed over its entire length ensuring even air distribution in the box-type body.

The fuel for the heater is obtained via an ascending pipe from the vehicle tank. The heater system has its own delivery and feed pump. The combustion air is obtained from the ventilated area in which the heater is installed.

The exhaust gases are directed by means of an exhaust pipe through a hole in the body floor to the outside. The exhaust pipe ends between the two side members of the frame with the exhaust gas outlet opening facing to the rear. If necessary, thermal insulation can be provided within the area in which the heater is installed. The hole in the floor of the body is adequately sized so that it also serves as a ventilation hole for the heater compartment.

A float switch is installed on the front left cross member of the platform. If the water level increases up to the float when driving through water, the overheating cutout responds and the heater is switched off. The overheating cutout must be actuated on the heater switch panel in order to switch on the heater again after driving through water. The float switch is protected from soiling by means of a sieve filter.

MBMANUALS.COM

Technical Safety and Operating Regulations

- The heater must not be in operation during refuelling and in closed rooms.
- Switch off heater at least three minutes before approaching a gasoline station, loading and unloading points.
- Disconnect auxiliary batteries when carrying out all work on the heater.
- The heat exchanger of the heater requires changing after 10 years and must be replaced by the manufacturer or one of his authorized workshops. A plate must then be attached to the heater indicating the date of purchase of the heat exchanger and the words "genuine spare part".

Troubleshooting

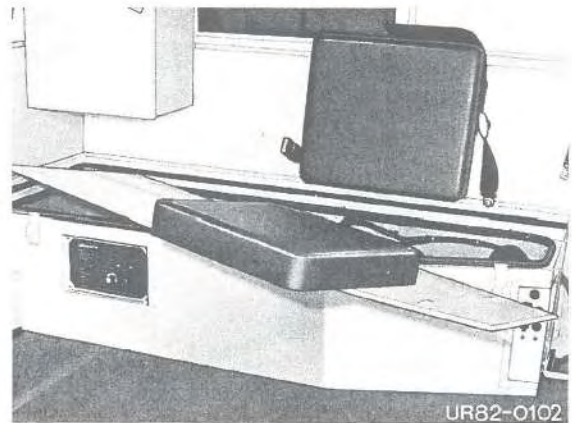
Fault	Cause	Remedy
Heater does not ignite, indicator lamp lights up and switches off again after 3 minutes	When switching on the heater, battery voltage between terminal 30 and 31 on heater control unit below 20 V	Start engine or charge battery
	"Heater- ventilator" switch defective	Check switch contacts, renew switch if necessary
	Relay 2 in switch box does energize when on switch "heater-ventilator" is in "heater" position	Check winding ($305 \pm 15 \Omega$), renew if necessary
	Heater control unit faulty	Check by exchanging
	No fuel in tank	Refuel, if the fuel line is empty, start heater 2-3 times to provide ventilation
	Overheating cutout has triggered	Actuate overheating cutout
	Overheating switch does not open	Check, renew overheating switch if necessary
	Glow plug defective	Check, renew if necessary
	Heater fan motor not running	Checking by applying the battery voltage to terminals 324 (+) and 31 H (-) via an 8 amp fuse. Repair burner if necessary, ensure the impellers are running freely
	Solenoid valve for combustion air does not open	Checking when removed: the solenoid must energize at 17 V, renew if necessary
After running of heater blower motor is not terminated automatically after normal switch-off	Pulse generator faulty	Check by exchanging
	Fuel quantity too great	Renew fuel feed pump if necessary
	Thermal switch faulty	Check by exchanging
Temperature of heated air too high	Heater control unit defective	Check by exchanging
	Faulty potentiometer in switch box	Check: set value $0-100 \Omega \pm 20 \Omega$, renew if necessary
	Relay 1 in switch box does not energize when turning rotary knob clockwise beyond the resistance felt	Exchange relay
	Temperature sensor in outlet scoop defective	Check: $108 \Omega \pm 2.5 \Omega$ at 20°C . Temperature -coefficient $TK = 0.6 \Omega/^\circ\text{C}$, renew if necessary

Troubleshooting

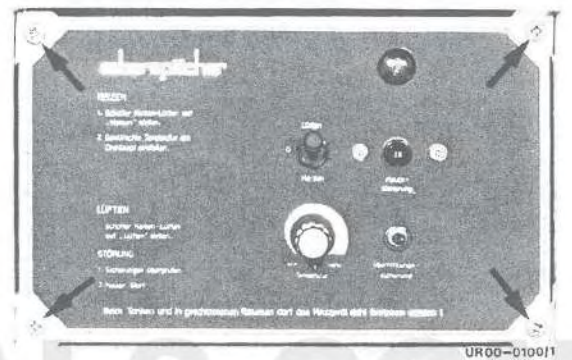
Fault	Cause	Remedy
Soot and smoke emission from heater (from the exhaust pipe)	Pulse generator defective	Check by exchanging
	Fuel quantity too great	Renew fuel feed pump if necessary
	Pulse generator defective	When the check of the burner has been completed as described above, check pulse generator by exchanging
	Heater control unit defective	Check by exchanging
	Atomizer in burner does not rotate although blower motor is running	Check by unscrewing the glow plug and observing through the plug hole. Renew coupling if necessary
	Fuel feed pump does not deliver fuel	Check whether the pulses of the pulse generator are applied at the terminals of the fuel feed pump. Renew pulse generator or feed pump if necessary
	Fuel feed pump is functioning but does not supply fuel	Check whether supply line is leaking, if necessary retighten hose clips, prefilter soiled, change filter (filter is seated in the union nut on the intake side)
	Heater control unit defective	If not already exchanged, check by exchanging
	No voltage at coil and ignition spark generator	Check, renew series resistor and noise suppression filter if necessary
	No ignition sparks	Check: disconnect high voltage connector and brown cable from glow plug. Hold high voltage connector against the connector on the brown cable (ground) at a distance of 5 mm. Renew ignition spark generator if necessary.
	Intake pipe for combustion air or exhaust pipe loose	Secure or exchange

Removal and Installation

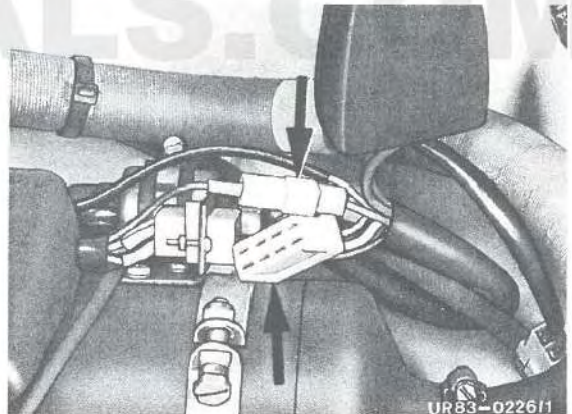
- 1 Remove cover plate.
- 2 Disconnect auxiliary batteries.



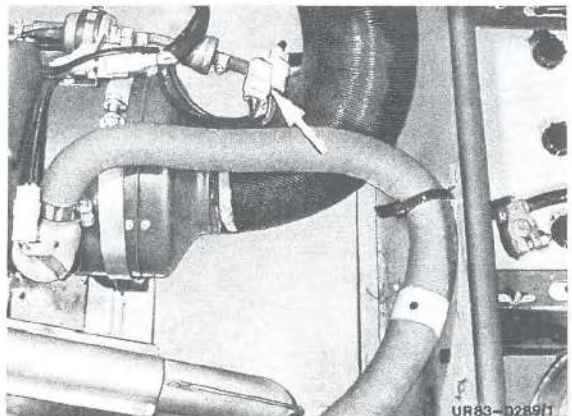
- 3 Unscrew switch panel and pull forward.



- 4 Disconnect main cable harness (8-pole) and cable colour code brown (1-pole).

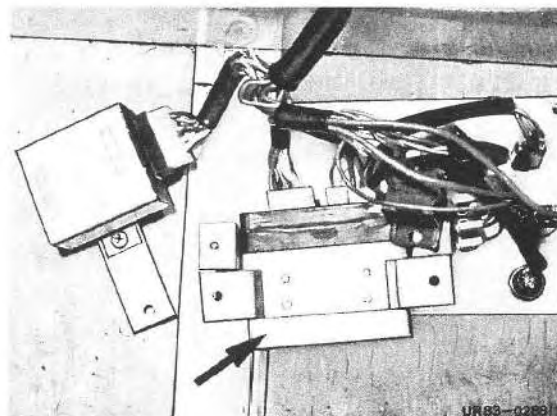
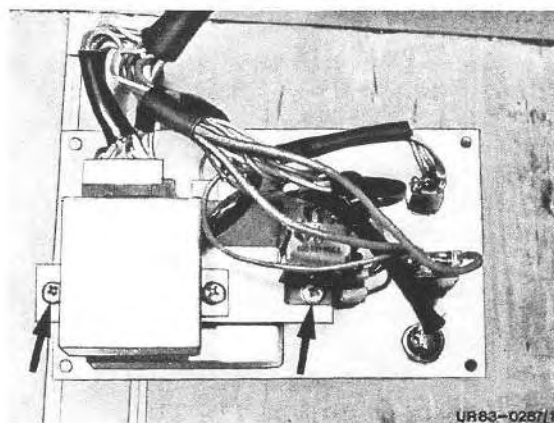
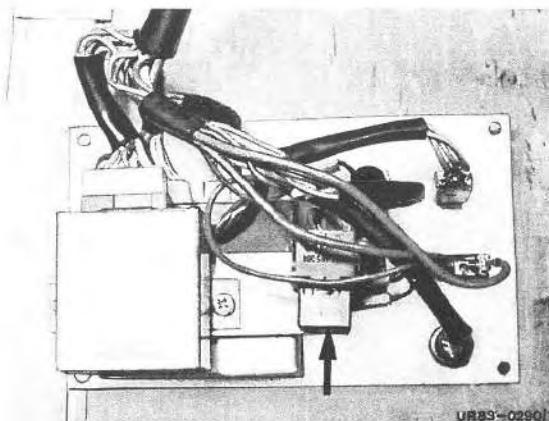


- 5 Disconnect cable colour code red/black (1-pole).
- 6 Disconnect electrical lines at the switch panel and remove switch panel.
- 7 Install in reverse order of removal.



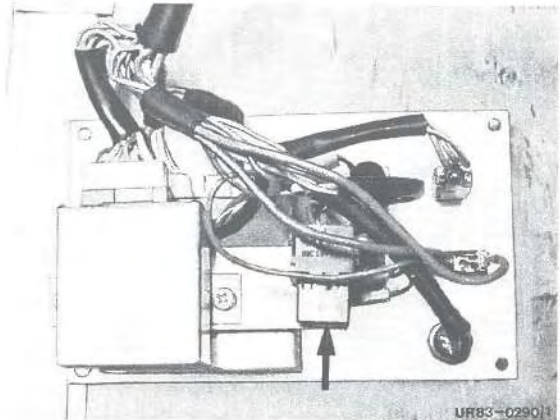
Removal and Installation

- 1 Remove switch panel, see 2.1/1.
- 2 Switch panel forward and tilt. Remove relay for float switch.
- 3 Remove bracket on pulse generator, place pulse generator to one side.
- 4 Take control unit out of bracket and remove electrical cable connectors.
- 5 Install in reverse order of removal
- 6 Carry out functional check.



Renewing

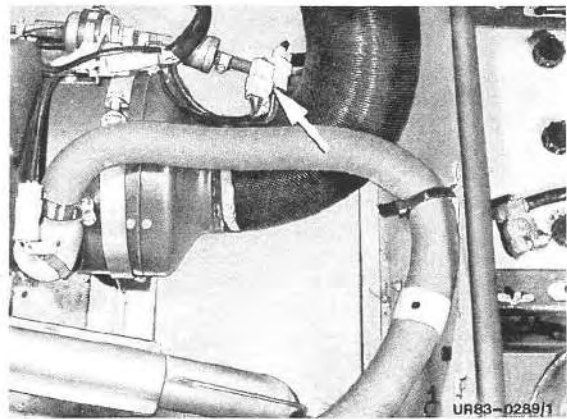
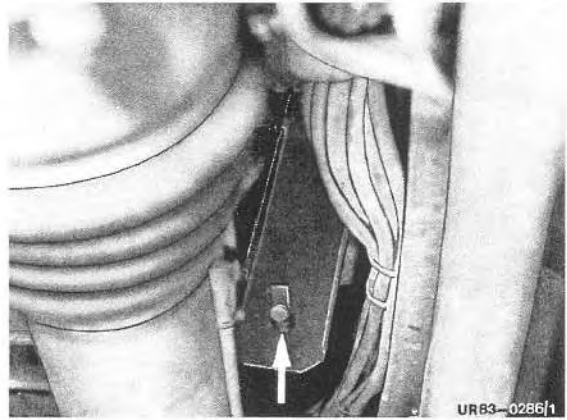
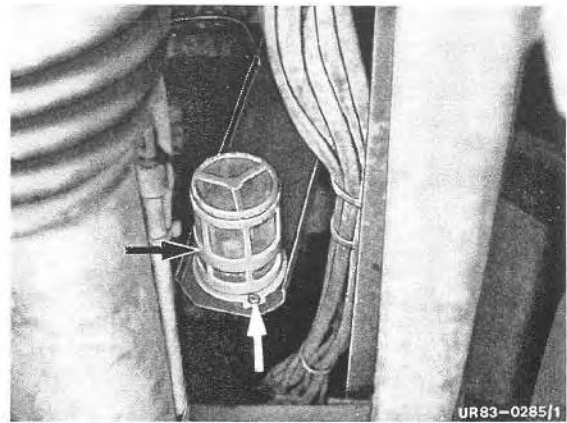
- 1 Remove switch panel, see 2.1/1.
- 2 Remove relay for float switch.
- 3 Install in reverse order of removal.



MBMANUALS.COM

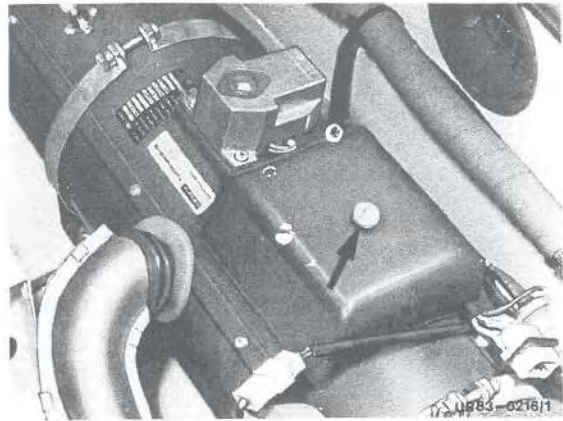
Removal and Installation

- 1 Remove cover plate.
- 2 Disconnect auxiliary batteries.
- 3 Unscrew sieve filter for float switch.
- 4 Pull float switch from bracket.
- 5 Detach electrical line from bracket and platform side member.
- 6 Disconnect electrical line (colour code red/black) at plug and socket connection in box-type body.
- 7 Remove electrical line.
- 8 Install in reverse order of removal.

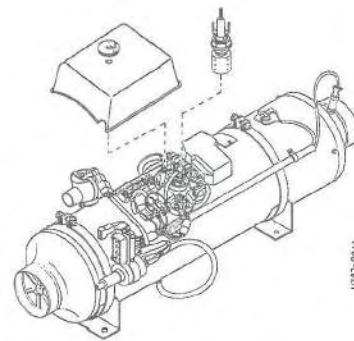


Renewing

- 1 Remove cover plate.
- 2 Disconnect auxiliary batteries.
- 3 Unscrew knurled nut and remove cover.
- 4 Release cables from glow plug and unscrew glow plug using special tool No. 1.

**Checking Function:**

- 1 Briefly connect coil to 12 V DC voltage supply. Core must glow uniformly, if necessary exchange glow plug.
- 2 Install in reverse order of removal.

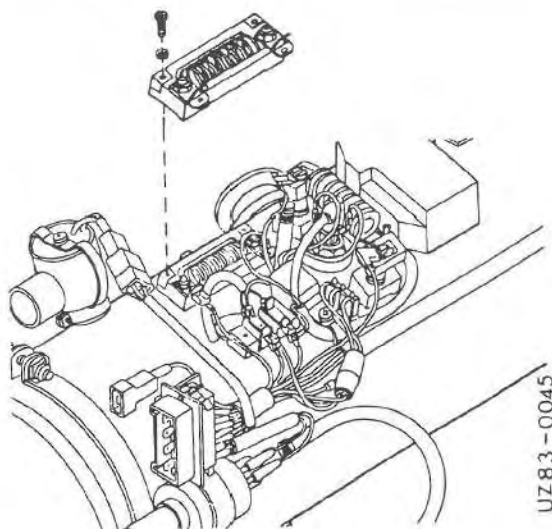


MBMANUALS.COM

Renewing

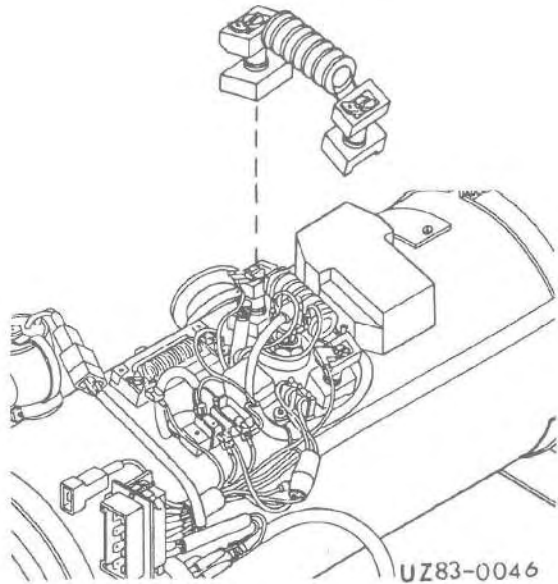
- 1 Remove cover plate.
- 2 Disconnect auxiliary batteries.
- 3 Remove cover.
- 4 Remove cable from series resistor.
- 5 Release 2 screws and exchange series resistor.

Note: The series resistor is installed only in ignition spark generators with a metal coloured housing.
No series resistor is installed in ignition spark generators with a blue plastic housing.



Renewing

- 1 Remove cover plate.
- 2 Disconnect auxiliary batteries.
- 3 Remove cover.
- 4 Disconnect cable from series resistor.
- 5 Unscrew 2 clamping screws and release fastening clip of the cable to ignition spark generator.
- 6 Pull series resistor out of holder and exchange.
- 7 Install in reverse order of removal.

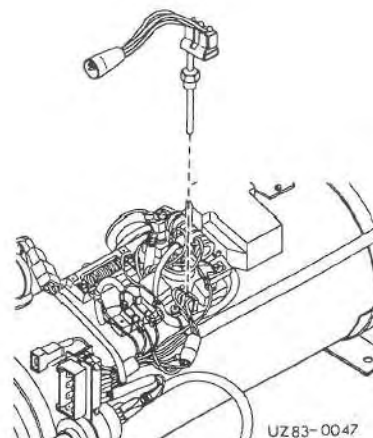


MBMANUALS.COM

Removal and Installation

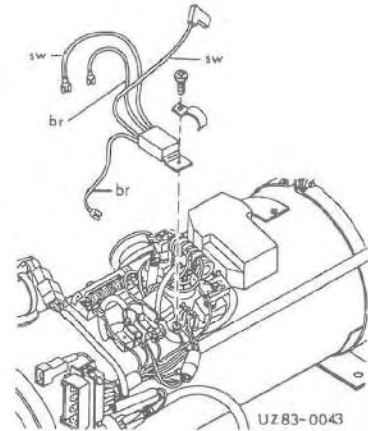
- 1 Remove cover plate.
- 2 Disconnect auxiliary batteries.
- 3 Remove cover.
- 4 Release electric connection and unscrew union nut.
- 5 Withdraw thermal switch and remove.
- 6 Install in reverse order of removal.

Installation note: Insert sensor tube in hole as far as it will go and tighten union nut together with sleeve.



Removal and Installation

- 1 Remove cover plate.
- 2 Disconnect auxiliary batteries.
- 3 Remove cover.
- 4 Unscrew fastening clip and disconnect plug-and-socket connections.
Black cable: from plug resistor
Brown cable: from distributor strip
Brown and black cable: from glow plug
- 5 Remove suppressor combination.
- 6 Install in reverse order of removal.



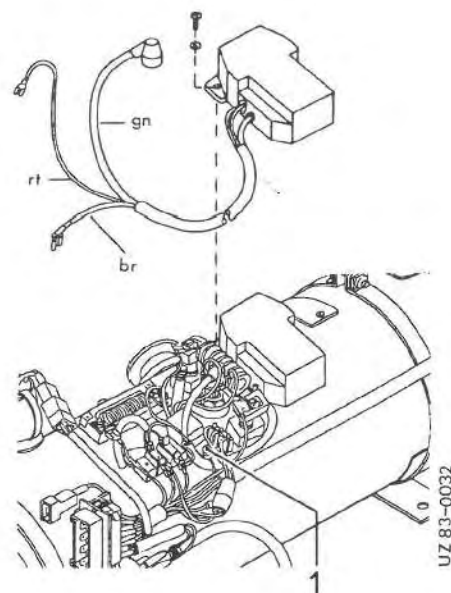
br= brown
sw= black

MBMANUALS.COM

Removal and Installation

- 1 Remove cover plate.
- 2 Disconnect (-) terminal of auxiliary batteries and place battery cable to one side.
- 3 Remove cover.
- 4 Unscrew fastening clip (1), disconnect plug-and-socket connection.
Brown cable: from distributor strip
Red cable: from series resistor for ignition spark generator
Green cable: from glow plug
- 5 Unscrew 2 screws, remove ignition spark generator and exchange.
- 6 Install in reverse order of removal.

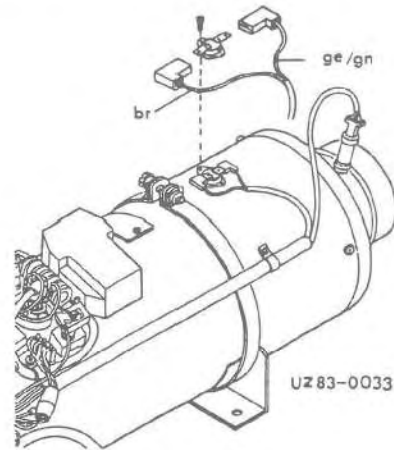
Note: The red cable for series resistor is installed only in ignition spark generators with a metal coloured housing. The series resistor is not required in ignition spark generators with a blue plastic housing.



br= brown
gn= green
rt= red

Removal and Installation

- 1 Remove cover plate.
- 2 Disconnect (-) terminal of auxiliary batteries and place battery cable to one side.
- 3 Remove connector housing, cables brown and yellow/green.
- 4 Unscrew overheating switch.
- 5 Install in reverse order of removal.



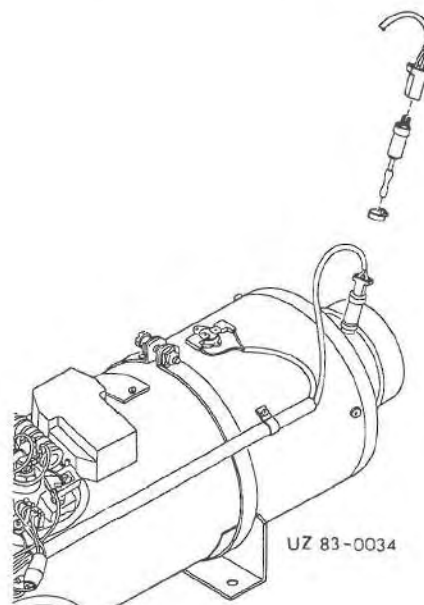
br= brown
ge= yellow
gn= green

MBMANUALS.COM

Removal and Installation

- 1 Disconnect electrical plug-and-socket connection.
- 2 Release clip, pull temperature sensor out of holder and remove.

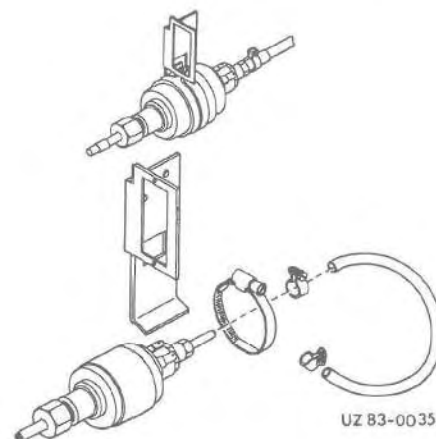
Note: Test values= $108 \Omega \pm 2,5 \Omega$ at 20°C .
Temperature coefficient $\text{TK} = 0,6 \Omega/^{\circ}\text{C}$.



Removal and Installation

- 1 Release clips from fuel hoses and remove fuel hoses.
- 2 Disconnect cables and release clip from fuel feed pump.
- 3 Withdraw feed pump, detach rubber ring and remove pump.
- 4 Install in reverse order of removal.

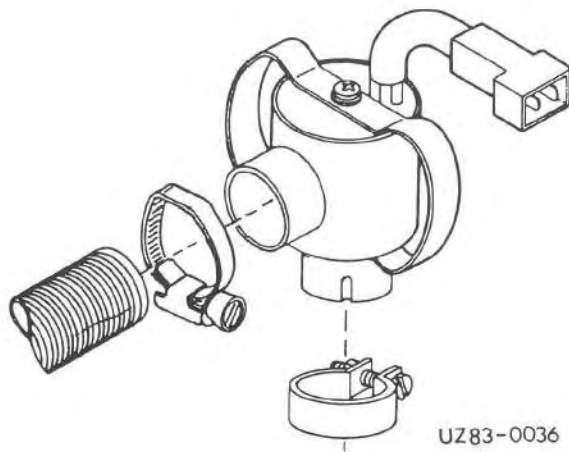
Installation note: After replacing the fuel feed pump, the entire fuel system is empty. The system is bled automatically by switching on the heater unit. If the safety time for the control unit (approx. 3 min.) is not sufficient for this purpose, the starting procedure should be repeated by switching off and on again.



MBMANUALS.COM

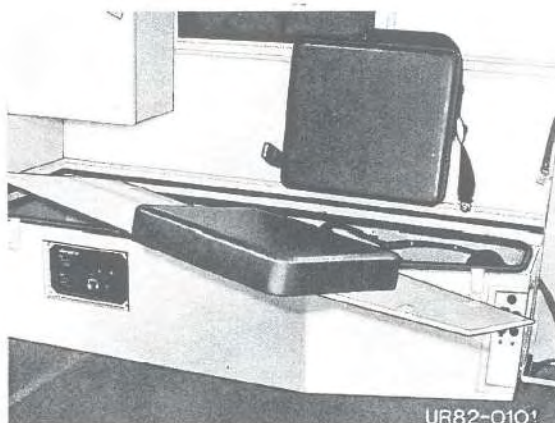
Renewing

- 1 Release 2 clips from solenoid valve, remove intake pipe from combustion air.
- 2 Disconnect electrical plug-and-socket connection.
- 3 Remove solenoid valve from socket and clean, remove if necessary.

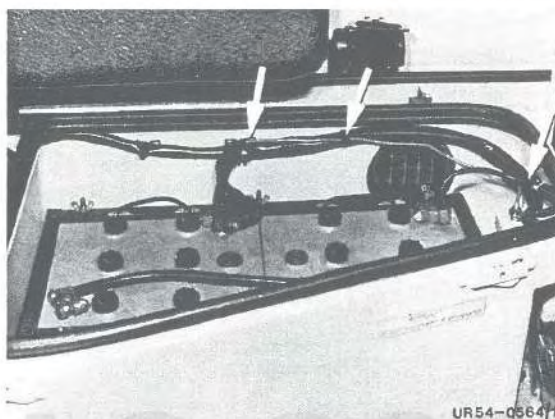


Removal and Installation

1 Remove cover plate.

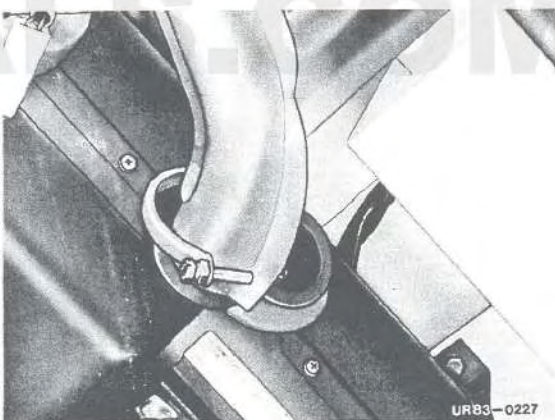


2 Disconnect auxiliary batteries.

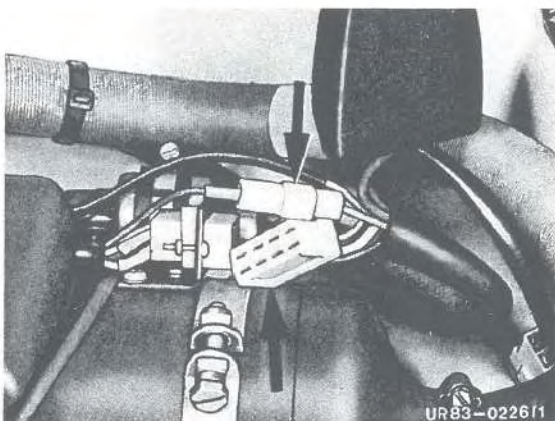


3 Release and remove exhaust line.

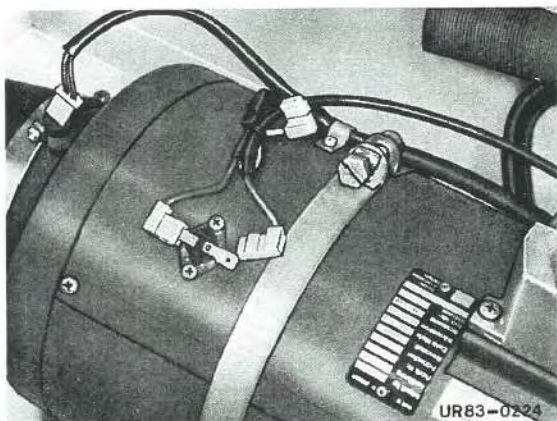
Note: Completely remove special clip. Pay attention to gasket!



4 Disconnect cable (8-pole) and cable (1-pole) at connecting point.

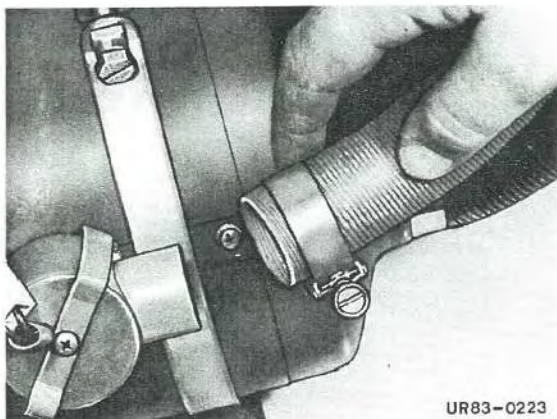


83.11 Removal and Installation of Heater Unit

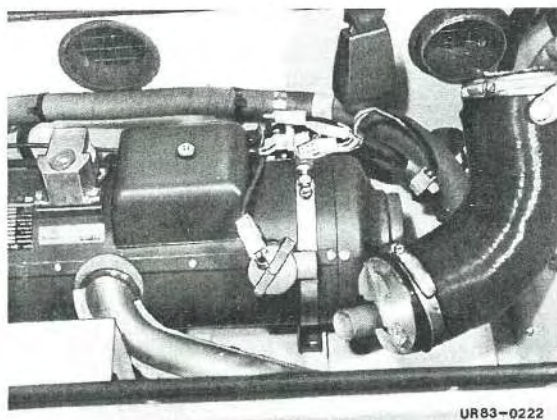


5 Cable at overheating switch, colour-code yellow/green.

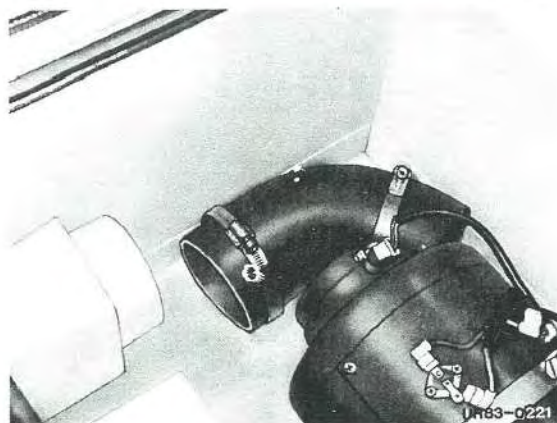
Note: 2. Cable with colour code yellow/green remains a dummy cable.



6 Release combustion air hose at the heater unit.



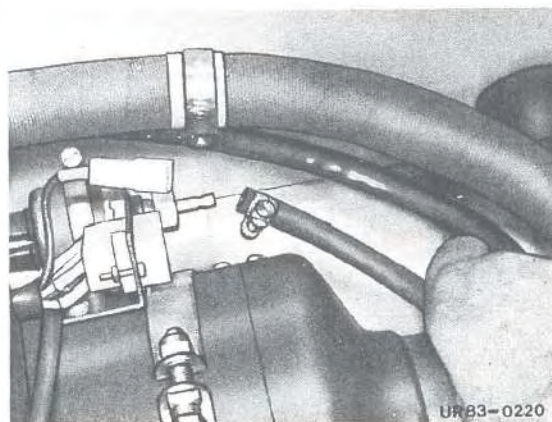
7 Release intake air hose at the pipe socket and heater unit and remove.



8 Release and remove heater hose at pipe socket and heater unit.

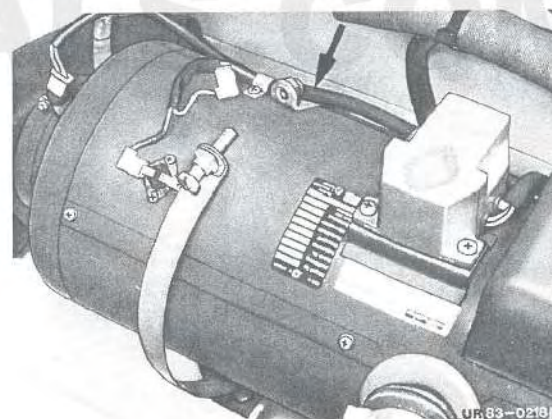
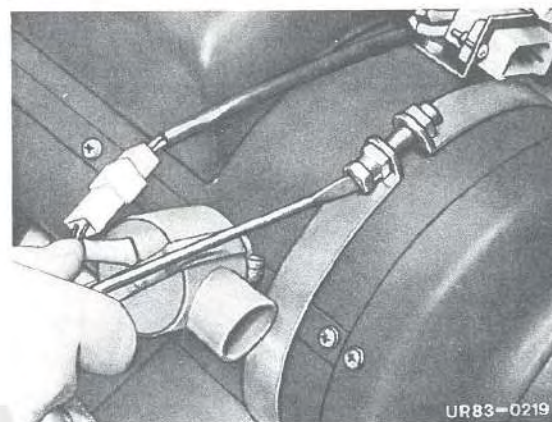
9 Release fuel line at the fuel pump and remove.

Note: Collect remaining fuel!

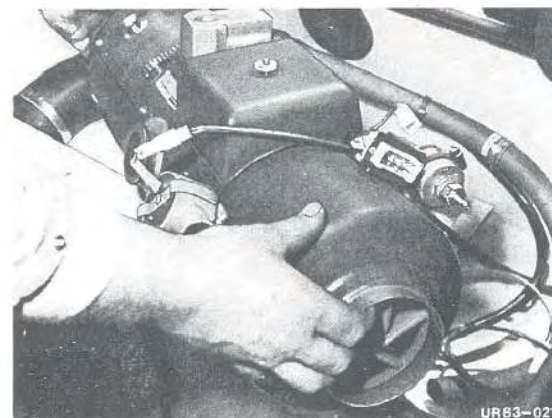


10 Release clip for securing the heater unit to the body.

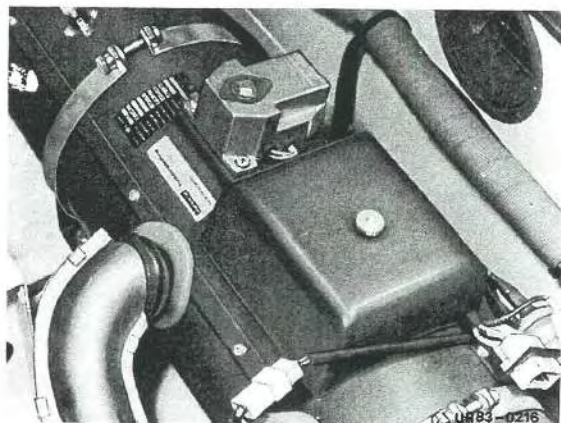
Note: When installing do not jam cable harness under clip.



12 Remove heater unit from heater compartment.

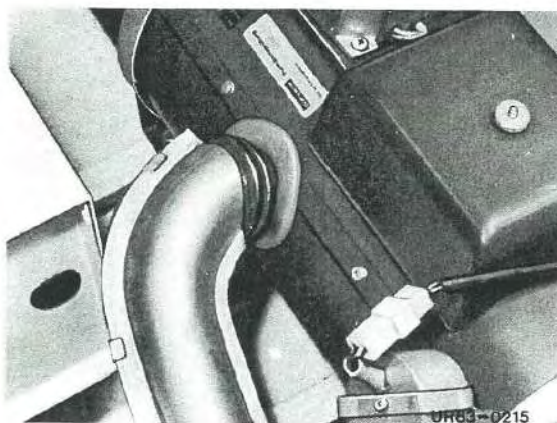


83.11 Removal and Installation of Heater Unit



13 Install in reverse order of removal.

Installation note: Before tightening the clips align heater unit relative to the exhaust line.

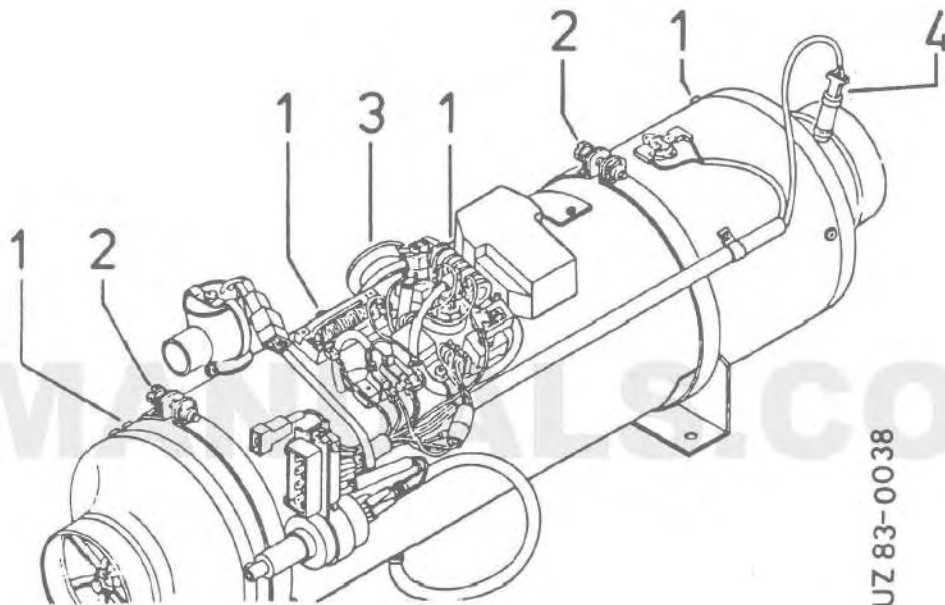


14 Tighten exhaust line, then secure heater unit with both retaining straps.

Note: Pay attention to gasket in the exhaust pipe.

Renewing

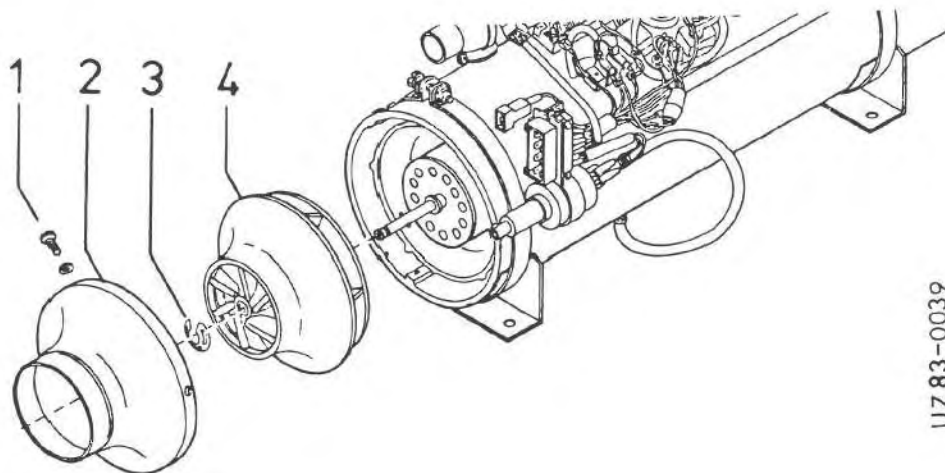
- 1 Remove heater unit, see 16.1/1.
- 2 Detach cover.
- 3 Disconnect electrical plug-and-socket connection from the temperature sensor (4).
- 4 Unscrew screws (1) from the outer jacket.
- 5 Release fastening clips (2).
- 6 Bend outer jacket by approx. 70 mm and cut through the old gasket (3) and remove.
- 7 Fit new silicone gasket on exhaust pipe socket.
- 8 Assemble in reverse order of removal.



UZ 83-0038

Renewing

- 1 Remove heater unit, see 16.1/1.
- 2 Unscrew 3 screws (1) from the inflow scoop.
- 3 Remove inflow scoop (2).
- 4 Remove retainer (3) from the shaft.
- 5 Remove and exchange impeller (4).
- 6 Install in reverse order of removal.



UZ 83-0039

Renewing

1 Remove heater unit, see 16.1/1.

2 Remove impeller, see 18.1/1.

3 Lift out rubber sleeve with screwdriver, disconnect motor cable plug connector (6) from the distributor strip and pull back motor cable through the opening.

4 Unscrew 2 screws (1), remove motor (3) and exchange.

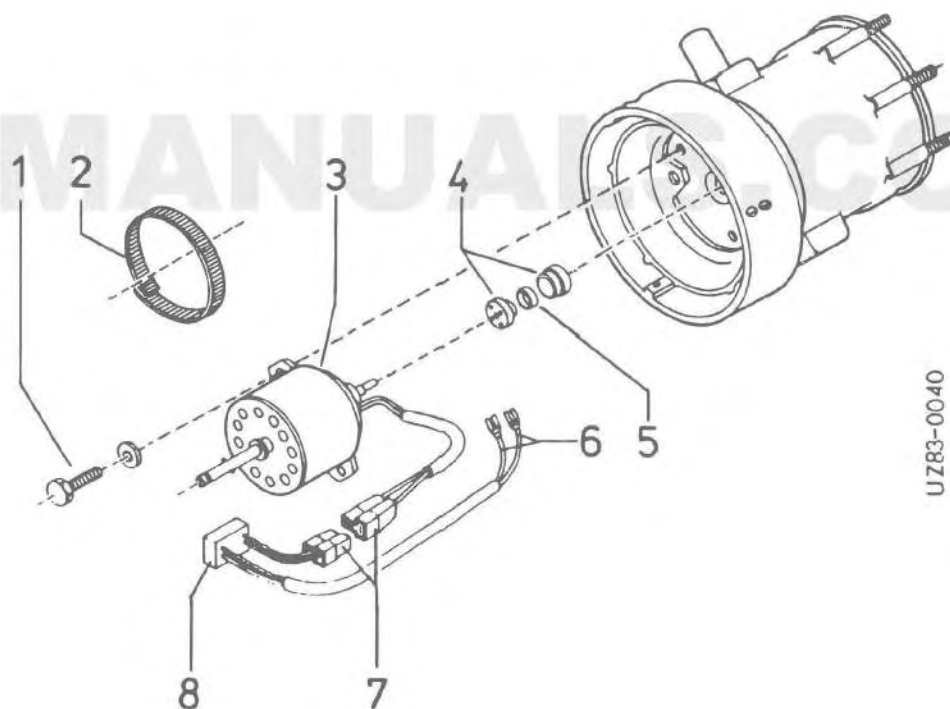
5 Release cable clip (2) and plug-and-socket connection (7), remove noise suppression filter (8) and exchange.

6 Remove coupling halves (4) from the shaft end and fit new coupling halves.

7 Insert driver plate (5) in clutch half.

Installation note: Fit motor (coupling finger must engage in driver plate), fasten motor with screws. (The insulating tube of the motor must not be fitted near the motor flange. The cables of the noise suppression filter must lie freely in the housing groove.) Coat motor shaft with anticorrosion agent.

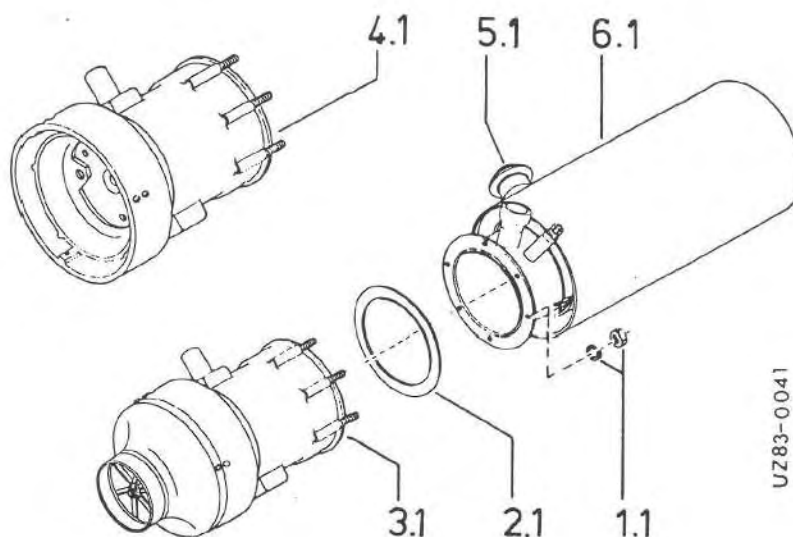
8 Further assembly in reverse sequence.



Renewing

- 1 Remove heater unit, see 16.1/1.
- 2 Detach cover.
- 3 Disconnect electrical connection (6) from the temperature sensor (9), see 20.1/3.
- 4 Detach solenoid valve (20), see 15.1/1.
- 5 Remove glow plug (2), see 6.1/1.
- 6 Remove thermal switch (3), see 9.1/1.
- 7 Lift out rubber sleeve (15) with screwdriver.
- 8 Disconnect motor cable plug-and-socket connections from the distributor strip (14) and pull back motor cable through the opening.
- 9 Release clip from the fuel hose, remove hose and lift out rubber sleeve (16) using screwdriver.
- 10 Release and remove 2 clips (7).
- 11 Unscrew 8 screws from the outer jacket.
- 12 Remove outflow scoop (8).
- 13 Carefully place unit on inflow scoop (26).
- Note:** Scoop must not be pushed in, otherwise it will catch on the impeller.
- 14 Bend outer jacket by approx. 70 mm and remove.
- 15 Unscrew 6 nuts (1.1).
- 16 Place 2 screwdrivers between flange of heat exchanger (6.1) and burner (3.1) and lift off heat exchanger.
- 17 Renew heat exchanger (6.1).
- 18 Exchange gasket (2.1).
- 19 Fit new silicone gasket (5.1) over exhaust pipe socket.
- 20 Remove inflow scoop and impeller see 18.1/1.
- 21 Remove electric motor and noise suppression filter see 19.1/1.
- 22 Exchange burner (4.1).

Note: Test run motor for short period.

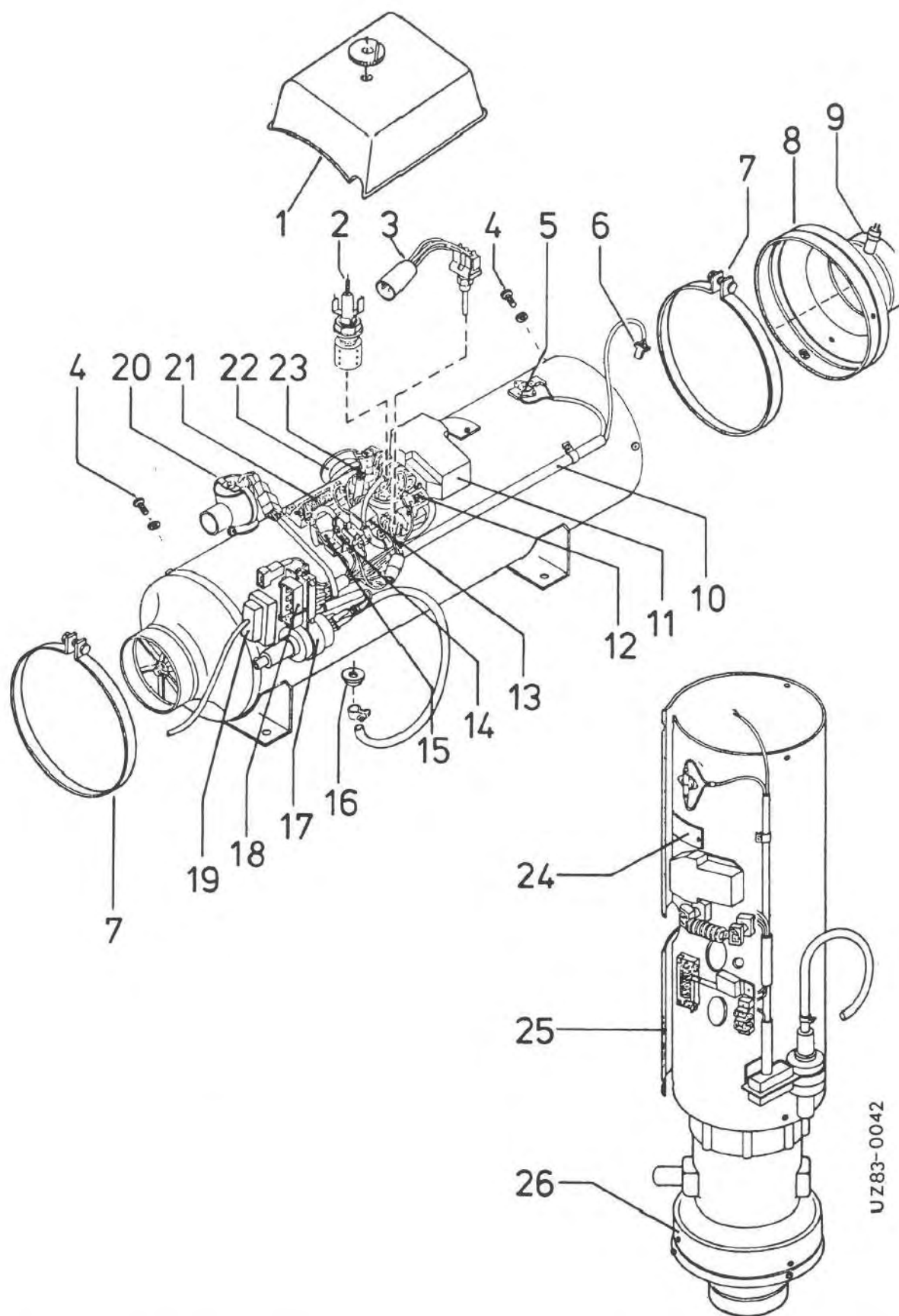


83.11 Renewing Heat Exchanger and Burner

Keys to 20.1/3

- | | |
|-----------------------------|---|
| 1 Cover | 14 Distributor strip |
| 2 Glow plug | 15 Rubber sleeve |
| 3 Thermal switch | 16 Rubber sleeve |
| 4 Screws | 17 Fuel feed pump |
| 5 Overheating switch | 18 8-pin plug connector |
| 6 Electrical connection | 19 Plug-and-socket connection |
| 7 Clamp | 20 Solenoid valve |
| 8 Outflow scoop | 21 Series resistor for ignition spark generator |
| 9 Temperature sensor | 22 Sealing ring |
| 10 Wire harness | 23 Stud |
| 11 Ignition spark generator | 24 Type plate |
| 12 Plug resistor | 25 Outer jacket |
| 13 Suppressor combination | 26 Inflow scoop |

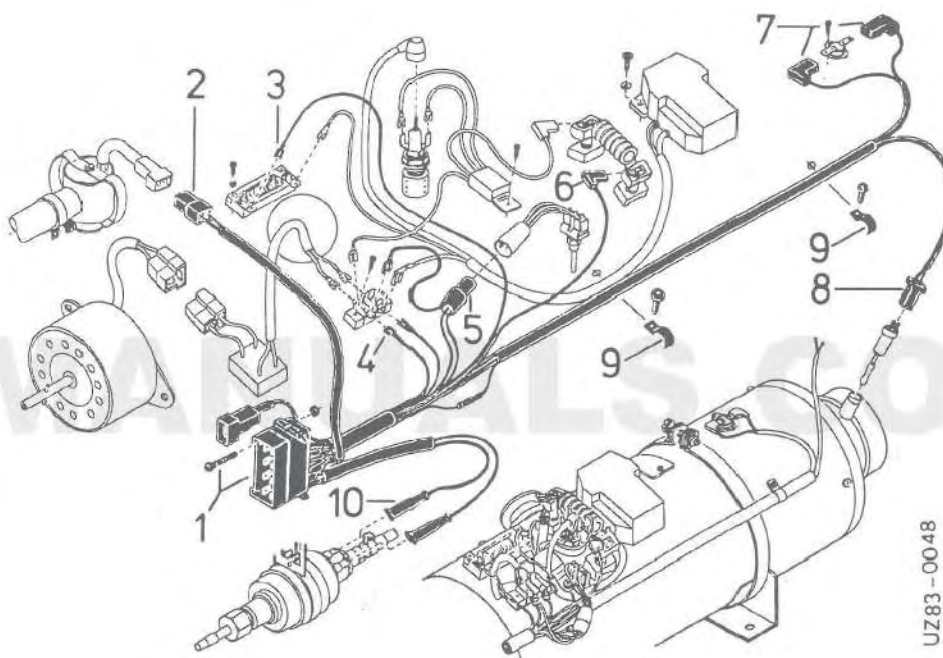
MBMANUALS.COM



UZ83-0042

Renewing

- 1 Remove heater unit, see 16.1/1.
- 2 Detach cover.
- 3 Unscrew screws of the 8-pin plug connector and pull plug connector (1) out of holder.
- 4 Disconnect plug connector plug-and-socket connections at the solenoid valve (2), series resistor for ignition spark generator (3), distributor strip (4), thermal switch (5) plug resistor (6), overheating switch (7), temperature sensor (8), and fuel feed pump (10).
- 5 Unscrew 2 fastening clips (9).
- 6 Remove and exchange cable harness.
- 7 Install in reverse order of removal.



General Note:

Required for measurement:

- 1 Dräger handpump (1)
- 2 Dräger tube (2) carbon oxide 10/a
- 3 Dräger tube (3) carbon oxide 0,1% a refer to page 22.1/2.

These parts can be obtained from a trade dealer.



UR00-0094

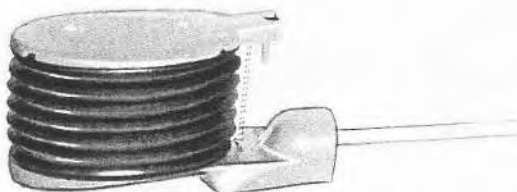
Checking

- 1 Close all doors, windows and hatches of the box-type body.
- 2 Switch on heater and set to maximum heat setting.
- 3 At an inner temperature between + 30° C to + 40° C start the CO value measurement using Dräger hand-pump with test tube— (carbon oxide 10/a— CH 28 900 —).
- 4 Carry out first measurement at the heater duct.

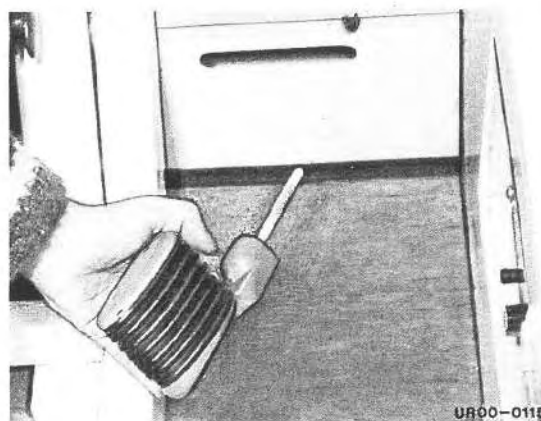
Note: Carry out measurement according to instruction for use for test tube CH 28 900 - Messrs. Dräger 17th edition August 1977.

- 5 Evaluate measured results.

Note: Maximum permissible work place concentration (MAK value - 50 ppm.)



UR00-0111



UR00-0112



UR00-0114

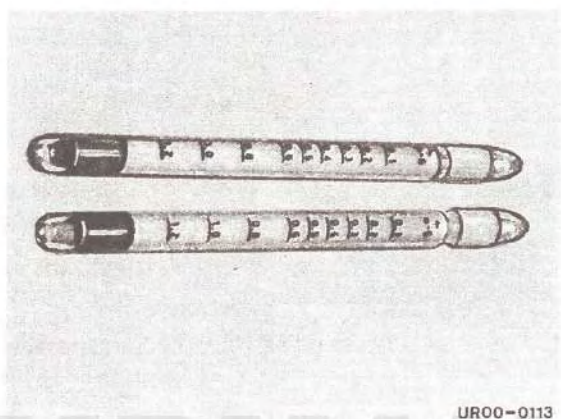
83.11 Checking CO Value and CO₂ Value in Interior of Box-Type Body



6 Carry out second measurement in the heater compartment and evaluate as described in points 4 and 5.

7 Carry out measurement and analysis of the CO₂ - values as described in points 1 to 6.

Note: For the CO₂ value measurement use test tube (3) carbon dioxide 0,1%/a - CH 23501 -
Permissible MAK value: 0,5 Vol%



MBMANUALS.COM